

Table 3 — Mechanistic Medicine, Clinical Decisions and Operational Optimization

BENQODHI Day 2 morning round tables · 11 September 2026

Printable guide for the table. Keep it in front of you during the discussion.

Aim of the round tables

Collect focused expert input for a possible BENQODHI output (paper, report, roadmap or policy note). The central question is how important challenges in health and the life sciences can become **credible, reusable benchmarks** for classical, AI-based, hybrid and quantum algorithms.

The preferred focus is **optimization** problems, but the discussion is not restricted to them: inference, sampling, simulation, reconstruction and learning problems also count when they have a clear health or life-science motivation and a plausible route to benchmarkability.

We are **not** building the benchmark repository during the workshop. The goal is to identify which problems would make a future benchmark repository useful.

The discussion path

health or life-science challenge → computational bottleneck → candidate benchmark → available evidence and missing pieces → fair comparison of algorithms → next benchmark-building action

This table's role

Identify candidate mechanistic, clinical and operational problems that could become benchmarkable. First collect possible candidates (from the prompts below, the [problem brief](#), or your own expertise), then select the strongest ones and answer the questions for each.

Problem prompts for discussion

Starting points only — replace or add to them when stronger candidates emerge.

- radiotherapy treatment planning;
- mechanistic-modelling problems such as network reconstruction, context-specific model extraction, parameter inference, model calibration and intervention search;
- healthcare scheduling, routing and resource allocation;
- molecular docking or other biomedical/pharmaceutical graph-optimization problems;

- simulation-based optimization where objective evaluations are costly, stochastic or model-dependent.

Focus: identify which mechanistic, clinical or operational problems preserve real health meaning while remaining precise enough to benchmark.

Questions to answer

For each selected candidate problem (aim for 2-3 candidates)

1. **Problem** — What health or life-science problem should be benchmarked, and why does it matter?
2. **Bottleneck** — What is hard computationally?
3. **Moving ahead** — What is missing, and what is the next practical step?

Transversal question — answer once for the table

Sustaining the ecosystem (infrastructure, community, scientific governance):

BENQODHI only works if what this table produces feeds one connected, durable ecosystem rather than a silo. Of the three pillars — the shared **infrastructure** that hosts and maintains it, the **community** that keeps contributing, and the **scientific governance** that keeps it credible — how can this table best feed the third pillar, scientific governance? What would it take for the problems, data and baselines from this table to be curated, reviewed and trusted as credible science over time, and what governance role should this table's own community play in setting and upholding those standards?

Expected output

- a short list of selected candidate problems;
- for each candidate, short answers to the three questions above;
- one answer to the transversal governance question.

Note takers: draft answers live into `answers.md` in this folder. Rapporteur: present the main conclusion, main candidate problem, why it matters, the most important missing piece, the next action, and the governance answer.